

Exercice 3.4

7) $y = \sin 4x$

$$u = 4x \quad y' = \cos(u) (4)$$
$$y = \sin(u) \quad y' = \cos(4x) (4) = \underline{4 \cos 4x}$$

9) $f(x) = \sqrt{5x+1}$

$$u = 5x+1 \quad f'(x) = \frac{1}{2\sqrt{u}} (5) = \frac{1}{2\sqrt{5x+1}} (5) = \underline{\frac{5}{2\sqrt{5x+1}}}$$
$$y = \sqrt{u}$$

11) $f(\theta) = \cos(\theta^2)$

$$u = \theta^2 \quad f'(\theta) = -\sin(u) (2\theta) = -\sin(\theta^2) (2\theta)$$
$$y = \cos(u) \quad f'(\theta) = -2\theta \sin \theta^2$$

17) $f(x) = (2x-3)^4 (x^2+x+1)^5$

$$p' = 4(2x-3)^3 (2)$$
$$s' = 5(x^2+x+1)^4 (2x+1)$$

$$f'(x) = (2x-3)^4 5(x^2+x+1)^4 (2x+1) + (x^2+x+1)^5 4(2x-3)^3 (2)$$

$$f'(x) = (2x-3)^3 (x^2+x+1)^4 [5(2x-3)(2x+1) + 8(x^2+x+1)]$$

$$f'(x) = (2x-3)^3 (x^2+x+1)^4 [(10x-15)(2x+1) + (8x^2+8x+8)]$$

$$f'(x) = (2x-3)^3 (x^2+x+1)^4 [20x^2+70x-30x-15+8x^2+8x+8]$$

$$f'(x) = (2x-3)^3 (x^2+x+1)^4 \underline{(28x^2-12x-7)}$$

21) $y = \sqrt{\frac{x}{x+1}}$

$$u = \frac{x}{x+1} \quad y' = \frac{1}{2\sqrt{u}} \left(\frac{1(x+1) - x(1)}{(x+1)^2} \right) = \frac{1}{2\sqrt{u}} \left(\frac{(x+1) - x}{(x+1)^2} \right)$$
$$y = \sqrt{u}$$

$$y' = \frac{1}{2\sqrt{\frac{x}{x+1}}} \left(\frac{(x+1) - x}{(x+1)^2} \right) = \underline{\frac{1}{2\sqrt{x}(x+1)^{3/2}}}$$

Scribe

$$13) y = e^{\tan \theta}$$

$$y' = e^{\tan \theta} (\sec^2 \theta) = (\sec^2 \theta) e^{\tan \theta}$$

$$29) H(r) = \frac{(r^2 - 1)^3}{(2r + 1)^5}$$

$$H'(r) = \frac{(2r+1)^5 \cdot 3(r^2-1)^2(2r) - (r^2-1)^3 \cdot 5(2r+1)^4 \cdot 2}{(2r+1)^{10}}$$

$$H'(r) = \frac{2(2r+1)^4 (r^2-1)^2 [3r(2r+1) - 5(r^2-1)]}{(2r+1)^{10}}$$

$$H'(r) = \frac{2(r^2-1)^2 [3r(2r+1) - 5(r^2-1)]}{(2r+1)^6}$$

$$35) y = \sin(\tan 2x)$$

$$u = \tan 2x$$

$$y = \sin(u)$$

$$y' = \cos(u) (\sec^2 2x) = \cos(\tan 2x) (\sec^2 2x)$$

$$y' = 2 \cos(\tan 2x) \sec^2 2x$$

$$39) f(t) = \tan(\sec(\cos t))$$

$$u = \sec(\cos t)$$

$$y = \tan(u)$$

$$f'(t) = \sec^2 u \sec u \tan u - \sin t$$

$$f'(t) = -\sec^2(\sec(\cos t)) \sec(\cos t) \tan(\cos t) \sin t$$

$$43) g(x) = (2ra^x + n)^p$$

$$u = 2ra^x + n$$

$$y = u^p$$

$$g'(x) = p u^{p-1} \cdot 2r (\ln a) a^x$$

$$g'(x) = p(2ra^x + n)^{p-1} 2r (\ln a) a^x$$

Scribe

$$45) y = \cos \sqrt{\sin(\tan \pi x)}$$

$$u = \sqrt{\sin(\tan \pi x)} \quad y' = -\sin u \left(\frac{1}{2\sqrt{u}} \right) \cos(\tan \pi x) \sec^2 \pi x \pi$$

$$y = \cos u$$

$$a = \sin(\tan \pi x) \quad y' = -\sin(\sqrt{\sin(\tan \pi x)})' \left(\frac{1}{2\sqrt{\sin(\tan \pi x)}} \right) \cos(\tan \pi x) \sec^2 \pi x$$

$$b = \sqrt{a}$$

$$y' = \frac{-\pi \cos(\tan \pi x) \sec^2 \pi x \sin \sqrt{\sin(\tan \pi x)}}{2\sqrt{\sin(\tan \pi x)}}$$

$$47) y = \cos(\sin 3\theta)$$

$$u = \sin 3\theta \quad y' = -\sin u \cos 3\theta$$

$$y = \cos u \quad y' = -\sin(\sin 3\theta) \cos 3\theta = -3 \cos 3\theta \sin(\sin 3\theta)$$

$$y'' = -3(\cos 3\theta \sin(\sin 3\theta) + (\sin \sin 3\theta) \cos 3\theta)$$

$$y'' = -3(-\sin 3\theta \cdot 3 \cdot \sin(\sin 3\theta) + \cos(\sin 3\theta) \cdot 3 \cos 3\theta)$$

$$y'' = -3(-3 \sin^2 3\theta \sin(\sin 3\theta) + 3 \cos^2 3\theta \cos(\sin 3\theta))$$

$$y'' = 9 \sin^2 3\theta \sin(\sin 3\theta) - 9 \cos^2 3\theta \cos(\sin 3\theta)$$

$$49) y = \sqrt{7 - \sec t}$$

$$u = 7 - \sec t \quad y' = \frac{1}{2\sqrt{u}} - \sec t \tan t = y' = \frac{-\sec t \tan t}{2\sqrt{7 - \sec t}}$$

$$y = \sqrt{u}$$

$$y'' = \frac{-\frac{1}{2} \left(\frac{(\sec t + 2 \sec^3 t) \sqrt{7 - \sec t}}{(7 - \sec t)^2} - \left(\frac{-\sec t \tan t}{2\sqrt{7 - \sec t}} \right) \sec t \tan t \right)}{1}$$

$$y'' = \frac{2(1 - \sec t + 2 \sec^3 t)(7 - \sec t) + \sec^2 t \tan^2 t}{4(7 - \sec t) \sqrt{7 - \sec t}}$$

Scribe

$$53) y = \sin(\sin x) \quad (\pi, 0)$$

$$u = \sin x$$

$$y = \sin u$$

$$y' = \cos u \cos x = \cos(\sin x) \cos x$$

$$y'|_{\pi} = \cos(\sin \pi) \cos \pi = -1$$

$$y - 0 = -1(x - \pi)$$

$$y = \underline{-x + \pi}$$